# Development of a Push-Tolerant Autonomous Blimp: Recovery After Physical Human Contact

Benjamin Greenberg[1], Prof. David Saldaña[2]
[1]Swarthmore College, [2]Lehigh University

## INTRODUCTION

- Quadrotors are loud, fly for under 8 minutes, and their exposed rotors make any contact hazardous.
- A helium blimp is quiet, flies far longer, and is safe to touch. [3]
- Contact becomes a routine disturbance, not an unusual occurrence.

## OBJECTIVE

- Build a blimp that returns to a commanded point after a hand push.
- Measure how far it is pushed, and how often and how quickly it recovers, with no operator input.
- Identify what limits how precisely it returns to the point.

## PLATFORM

- XIAO ESP32-S3 running ported Crazyflie firmware, carried in a 3D-printed gondola under a mylar envelope. [2, 4]
- Two forward motors steer the blimp differentially, while one up motor and one down motor hold its altitude.
- Each motor can only spin one way, so the blimp cannot reverse, brake, or turn in place, making minimum turn radius 1.1 to 1.4 m.
- A second ESP32 on USB bridges the motion capture network to the vehicle over ESP-NOW radio.

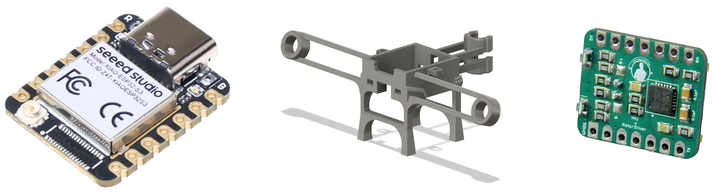



**Figure 2.** ESP32 flight board, printed gondola frame, and motor driver with IMU.



**Figure 1.** The blimp: mylar envelope, printed gondola, four brushed motors.

## METHODS

- All guidance runs on blimp, station streams pose and gains only.
- Steering follows pure pursuit, tracking a carrot placed a fixed lookahead along the path, with a PD controller using gyroscope. [1]

$$u_{turn} = k_p \cdot e_\psi - k_d \cdot r$$

$$u_{fwd} = F_{max} \cdot \max(\alpha, \cos e_\psi)$$

$e_\psi$ = heading error  $r$ = yaw rate  $\alpha$ = alignment floor

- Forward thrust is set on heading alignment, so it runs at full thrust only when aimed at the target.
- One waypoint set, the blimp converges on it, is pushed by hand, and then returns with no operator input.
- n = 23 disturbances using the final tuned configuration.

## DISCUSSION

- Stands in for a rehabilitation interaction, where the push is the exercise and the log is the record.
- Each push is a single impulse, so the vehicle is genuinely released.
- Constant cruise speed means it orbits rather than parks.
- Cruise speed sets recovery time: about 6.6 s per meter.

## CONCLUSIONS

- A soft, buoyant vehicle can treat a hand push as routine, and all 23 pushes recovered with no operator input.
- Alignment-gated thrust was enough to close on the point despite having no braking or reverse.
- The limit is precision, not recovery: without braking it orbits the point, so 78% settled within 0.8 m.

## FUTURE WORK

- Braking, through terminal deceleration and H-bridge drivers, so it can park at the point.
- An on-board camera, so it can fly not only in motion-capture spaces.
- Real rehabilitation trials with a person in the loop.

## ACKNOWLEDGEMENTS AND REFERENCES

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[1] Coulter. Implementation of the Pure Pursuit Path Tracking Algorithm. CMU-RI-TR-92-01, 1992.
[2] Xu, D'Antonio, Ammirato, Saldana. SBlimp: design, model and control of a swing-blimp. IROS, 2023.
[3] Yao et al. Autonomous flying blimp interaction with human in an indoor space. FITEE, 2019.
[4] Bitcraze, Crazyflie firmware documentation; Espressif, ESP-NOW user guide.

## RESULTS

**23 / 23** pushes returned toward the point

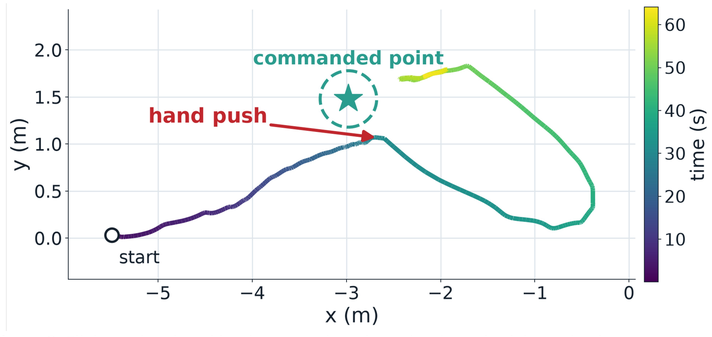



**Figure 3.** Flight path colored by time: acquires the point, is pushed, loops back.

**4.1 m** largest push

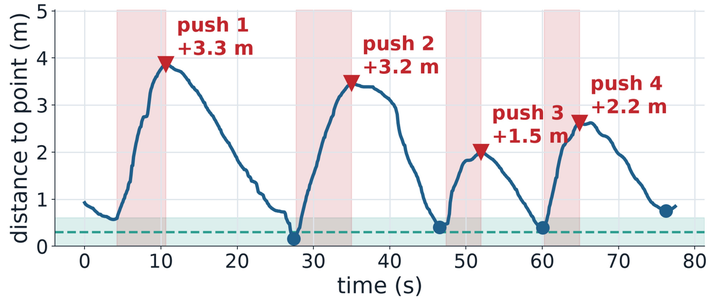



**Figure 4.** Four consecutive pushes of 1.5 to 3.3 m, each rejected in 8 to 17 s.

**78%** settled within 0.8 m

**15 s** median time to remove 80% of the offset

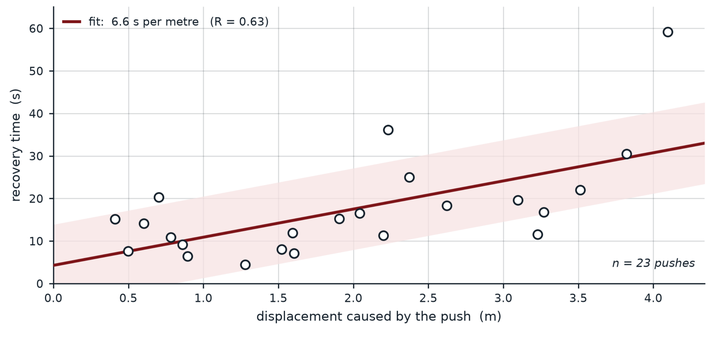



**Figure 5.** Recovery time against push displacement, all 23 disturbances.